

Prove global convergence to the physical invariant state for hydrodynamic equations

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Website #17 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Fix $d \geq 2$, $\lambda \in (0, 1)$, and a service distribution G with density g , tail $\bar{G}$, right endpoint $L \in (0, \infty]$, hazard rate $h = g/\bar{G}$, and mean 1. Let $\nu(t) = (\nu_\ell(t))_{\ell \geq 1} \in C_S[0, \infty)$ be the unique solution of the SQ(d) hydrodynamic equations with initial condition $\nu(0) \in \mathcal{S}$, where $\nu_\ell(t)$ is the finite measure of service ages at queues of length at least ℓ . Assume the hydrodynamic equations are well posed and that there is a unique physical invariant state $\nu^* \in \mathcal{S}$, i.e. a unique invariant state with $\langle 1, \nu_\ell^* \rangle \rightarrow 0$.

The target statement is the large-time convergence

$$\nu(t) \xrightarrow[t \rightarrow \infty]{} \nu^*$$

in an appropriate topology on $\mathcal{S}$.

The strongest literal reading would ask for this convergence for *every* $\nu(0) \in \mathcal{S}$. The negation of that strongest reading is: there exists $\nu(0) \in \mathcal{S}$, an $\varepsilon > 0$, a coordinate ℓ , and a bounded continuous f such that

$$|\langle f, \nu_\ell(t_n) \rangle - \langle f, \nu_\ell^* \rangle| \geq \varepsilon \quad \text{for some sequence } t_n \rightarrow \infty.$$

What I will prove is a rigorous asymptotic-stability theorem for two substantial subclasses of initial conditions/regimes:

- empty initial condition, in the *small-load* regime covered by Bramson–Lu–Prabhakar;
- an *order interval of initial conditions* between the empty law and the equilibrium environment, in the FIFO+DHR regime.

These are genuine long-time convergence results for the present SQ(d) hydrodynamic equations.

Alternative readings considered

- **Reading A (too strong):** convergence for all $\nu(0) \in \mathcal{S}$. This is false: the 2025 invariant-state paper explicitly notes a nonphysical invariant state with $s_\ell \equiv 1$ for all ℓ , so no convergence-to- ν^* theorem can hold on all of $\mathcal{S}$. (par.nsf.gov)
- **Reading B (the meaningful one):** convergence for a large physically relevant subclass of initial conditions. This is the reading I adopt.
- **Topology:** I use the product weak topology on $\mathcal{S}$: $\nu(t) \rightarrow \nu^*$ means that for every $\ell \geq 1$ and every bounded continuous $f : [0, L) \rightarrow \mathbb{R}$,

$$\langle f, \nu_\ell(t) \rangle \rightarrow \langle f, \nu_\ell^* \rangle.$$

Related literature found

- **Agarwal–Ramanan (2025), *Invariant States of Hydrodynamic Limits of Randomized Load-Balancing Networks*** — BACKGROUND/PARTIAL. This is the exact invariant-state paper behind the prompt. It characterizes invariant states and explicitly says that proving convergence of hydrodynamic solutions to the invariant state for a large class of initial conditions is future work; hence the full problem is not already settled there. (par.nsf.gov)
- **Aghajani–Ramanan (2019), *The hydrodynamic limit of a randomized load balancing network*** — TOOL. This supplies the mean-field limit: scaled state processes converge to the unique hydrodynamic solution, and finite-queue asymptotic independence is also established. I use only the hydrodynamic-limit part as a tool. (arxiv.org)
- **Bramson–Lu–Prabhakar (2012), *Asymptotic independence of queues under randomized load balancing*** — TOOL. In the FIFO+DHR regime, and also in a sufficiently small-load regime with finite second moment, this paper proves convergence of finite- N equilibrium marginals to a unique equilibrium environment; its Section 5/6 also gives N -uniform relaxation of the empty-start tagged queue to equilibrium. This is the key input for my successful argument. (web.stanford.edu)
- **Aghajani–Li–Ramanan (2017), *The PDE Method for the Analysis of Randomized Load Balancing Networks*** — BACKGROUND. Useful for cross-checking the interpretation of the state descriptor and the tagged-queue identities; not used as a main proof input. (dam.brown.edu)
- **Atar–Kang–Kaspi–Ramanan (2023), *Long-Time Limit of Nonlinearly Coupled Measure-Valued Equations that Model Many-Server Queues with Reneging*** — BACKGROUND. This is the closest conceptual neighbor: a large-time convergence theorem for a related coupled measure-valued queueing fluid model. It does *not* cover $SQ(d)$, but it informed the search for Lyapunov and renewal-based arguments. (epubs.siam.org)
- **Long–Zhang (2014), *Convergence to equilibrium states for fluid models of many-server queues with abandonment*** — BACKGROUND. Another related long-time result for many-server fluid models; methodologically suggestive but not directly applicable to the $SQ(d)$ hydrodynamic equations. (sciencedirect.com)

Proof sketch

A direct attack on the nonlinear measure-valued hydrodynamic equations did not yield a coercive Lyapunov functional. The successful idea is instead to *pull asymptotic stability from the prelimit*. The strategy has three steps.

First, I encode a one-queue law H on the queue-state space

$$\mathcal{Q} := \{0\} \sqcup \bigsqcup_{k \geq 1} (\{k\} \times [0, L))$$

into a hydrodynamic state $T(H) \in \mathcal{S}$ by

$$T(H)_\ell(A) := H(\{k \geq \ell\} \times A).$$

Thus $T(H)_\ell$ is exactly the age measure of queues of length at least ℓ .

Second, in the regimes treated by Bramson–Lu–Prabhakar, the *tagged queue* of the finite- N system started empty relaxes to the finite- N equilibrium *uniformly in N* . Combining that uniform relaxation with the hydrodynamic limit of Aghajani–Ramanan and the convergence $E^{(N,1)} \Rightarrow E^\infty$ of equilibrium marginals, I obtain convergence of the empty-start hydrodynamic solution to the invariant state induced by E^∞ , hence to ν^* .

Third, in the FIFO+DHR regime, Bramson–Lu–Prabhakar also provide an order-preserving coupling. By coupling three finite- N systems (empty, a general ordered initial law H_0 , and the equilibrium environment E^∞ sampled i.i.d. across queues), passing to the hydrodynamic limit, and then squeezing between the empty-start solution and the invariant state, I obtain convergence for every H_0 in the stochastic order interval

$$\delta_0 \preceq H_0 \preceq E^\infty.$$

So the final result is a rigorous partial asymptotic-stability theorem: empty-start convergence in the small-load regime, and convergence on a large monotone order interval in the FIFO+DHR regime.

Main result

Theorem 1. *Let $d \geq 2$ and $\lambda \in (0, 1)$. Assume:*

1. *the $SQ(d)$ hydrodynamic equations are well posed for all initial conditions in $\mathcal{S}$;*
2. *there is a unique physical invariant state $\nu^* \in \mathcal{S}$.*

For a probability law H on

$$\mathcal{Q} := \{0\} \sqcup \bigsqcup_{k \geq 1} (\{k\} \times [0, L]),$$

define $T(H) \in \mathcal{S}$ by

$$T(H)_\ell(A) := H(\{k \geq \ell\} \times A), \quad \ell \geq 1, \quad A \in \mathcal{B}([0, L]).$$

Then the following hold.

- (I) **FIFO+DHR ordered-initial-data convergence.** *Assume the service discipline is FIFO and h is decreasing. Let E^∞ be the unique equilibrium environment of Bramson–Lu–Prabhakar, and write $\bar{\nu} := T(E^\infty)$. Then $\bar{\nu}$ is a physical invariant state, hence $\bar{\nu} = \nu^*$ by uniqueness.*

If H_0 is any probability law on $\mathcal{Q}$ such that

$$\delta_0 \preceq H_0 \preceq E^\infty$$

in the queue-state stochastic order induced by

$$0 \preceq (k, a), \quad (k, a) \preceq (k', a') \iff k \leq k' \text{ and } a \leq a',$$

and if $\nu(\cdot)$ is the hydrodynamic solution with initial condition $\nu(0) = T(H_0)$, then

$$\nu(t) \xrightarrow[t \rightarrow \infty]{} \nu^*$$

in product weak topology; equivalently,

$$\langle f, \nu_\ell(t) \rangle \rightarrow \langle f, \nu_\ell^* \rangle \quad \text{for every } \ell \geq 1, \quad f \in C_b([0, L]).$$

(II) **Small-load empty-start convergence.** Assume G has finite second moment θ and

$$\lambda \leq \frac{1}{2\sqrt{d(d \vee \theta)}}.$$

Let $\nu^\emptyset(\cdot)$ be the hydrodynamic solution with empty initial condition

$$\nu_\ell^\emptyset(0) = 0, \quad \ell \geq 1.$$

Then

$$\nu^\emptyset(t) \xrightarrow[t \rightarrow \infty]{} \nu^*$$

in product weak topology.

Proof

I first set up the correspondence between one-queue laws and hydrodynamic states.

Lemma 1. Let H be a probability law on $\mathcal{Q}$. Define $T(H) = (T(H)_\ell)_{\ell \geq 1}$ by

$$T(H)_\ell(A) := H(\{k \geq \ell\} \times A).$$

Then $T(H) \in \mathcal{S}$.

Conversely, if $\mu = (\mu_\ell)_{\ell \geq 1} \in \mathcal{S}$ and $\mu_\ell([0, L]) \rightarrow 0$, then the formula

$$H_\mu(\{0\}) = 1 - \langle 1, \mu_1 \rangle, \quad H_\mu(\{k\} \times A) := (\mu_k - \mu_{k+1})(A), \quad k \geq 1,$$

defines a probability law on $\mathcal{Q}$, and $T(H_\mu) = \mu$.

Finally, for every bounded continuous $f : [0, L] \rightarrow \mathbb{R}$, if

$$\Phi_{\ell, f}(0) := 0, \quad \Phi_{\ell, f}(k, a) := f(a) \mathbf{1}_{\{k \geq \ell\}},$$

then

$$\int_{\mathcal{Q}} \Phi_{\ell, f} dH = \langle f, T(H)_\ell \rangle.$$

Proof. For $A \in \mathcal{B}([0, L])$,

$$T(H)_\ell(A) = H(\{k \geq \ell\} \times A) \geq H(\{k \geq \ell + 1\} \times A) = T(H)_{\ell+1}(A),$$

so $T(H) \in \mathcal{S}$.

For the converse, since $\mu_k \geq \mu_{k+1}$ as finite measures, each difference $\mu_k - \mu_{k+1}$ is a finite nonnegative measure. Also,

$$H_\mu(\{0\}) + \sum_{k \geq 1} H_\mu(\{k\} \times [0, L]) = 1 - \langle 1, \mu_1 \rangle + \sum_{k \geq 1} (\langle 1, \mu_k \rangle - \langle 1, \mu_{k+1} \rangle) = 1,$$

because $\langle 1, \mu_k \rangle \rightarrow 0$. Thus H_μ is a probability law. Moreover,

$$T(H_\mu)_\ell(A) = \sum_{k \geq \ell} (\mu_k - \mu_{k+1})(A) = \mu_\ell(A)$$

by telescoping and $\mu_k(A) \downarrow 0$ as $k \rightarrow \infty$ for every A , again because $\mu_k \leq \mu_k([0, L]) \rightarrow 0$.

The last identity is immediate:

$$\int_{\mathcal{Q}} \Phi_{\ell, f} dH = \sum_{k \geq 1} \int_{[0, L]} f(a) \mathbf{1}_{\{k \geq \ell\}} H(\{k\} \times da) = \int_{[0, L]} f(a) H(\{k \geq \ell\} \times da) = \langle f, T(H)_\ell \rangle. \quad \square$$

Lemma 2 (Empty-start convergence principle). *Assume there exists a probability law E^∞ on $\mathcal{Q}$ such that:*

(a) *for the finite- N equilibrium one-queue marginal $E^{(N,1)}$,*

$$E^{(N,1)} \Rightarrow E^\infty \quad (N \rightarrow \infty);$$

(b) *for every $\varepsilon > 0$ and every $T > 0$, there exists $q_1 = q_1(\varepsilon, T)$ such that, for every N and every $q \geq q_1$, one can couple the empty-start N -server system and the equilibrium-start N -server system so that queue 1 is identical throughout $[q, q + T]$ except on an event of probability at most ε ;*

(c) *the hydrodynamic limit theorem applies to the empty initial condition.*

Let $\bar{\nu} := T(E^\infty)$. Then the hydrodynamic solution $\nu^\emptyset(\cdot)$ started from the empty initial condition satisfies

$$\nu^\emptyset(t) \xrightarrow[t \rightarrow \infty]{} \bar{\nu}$$

in product weak topology.

Proof. Fix $\ell \geq 1$ and $f \in C_b([0, L])$. Let $\Phi_{\ell, f}$ be as above. Apply assumption (b) with $T = 1$. If $t \geq q_1(\varepsilon, 1) + 1$, choose $q = t - 1$. Then the tagged queue in the empty-start and equilibrium-start systems is equal at time t outside an event of probability at most ε . Therefore, for every N ,

$$\left| \mathbb{E}[\Phi_{\ell, f}(X_1^{N, \emptyset}(t))] - \mathbb{E}_{E^{(N,1)}}[\Phi_{\ell, f}] \right| \leq 2\|f\|_\infty \varepsilon. \quad (1)$$

Now pass $N \rightarrow \infty$. By exchangeability of the N -server system,

$$\mathbb{E}[\Phi_{\ell, f}(X_1^{N, \emptyset}(t))] = \mathbb{E}\langle f, \nu_\ell^{(N), \emptyset}(t) \rangle.$$

By the hydrodynamic limit theorem, $\nu^{(N), \emptyset}(t) \rightarrow \nu^\emptyset(t)$ for each fixed t , and the integrand is bounded by $\|f\|_\infty$, so

$$\lim_{N \rightarrow \infty} \mathbb{E}[\Phi_{\ell, f}(X_1^{N, \emptyset}(t))] = \langle f, \nu_\ell^\emptyset(t) \rangle. \quad (2)$$

By (a),

$$\lim_{N \rightarrow \infty} \mathbb{E}_{E^{(N,1)}}[\Phi_{\ell, f}] = \int_{\mathcal{Q}} \Phi_{\ell, f} dE^\infty = \langle f, \bar{\nu}_\ell \rangle. \quad (3)$$

Taking $N \rightarrow \infty$ in (1) and using (2)–(3),

$$|\langle f, \nu_\ell^\emptyset(t) \rangle - \langle f, \bar{\nu}_\ell \rangle| \leq 2\|f\|_\infty \varepsilon \quad \text{for all } t \geq q_1(\varepsilon, 1) + 1.$$

Since $\varepsilon > 0$ is arbitrary,

$$\langle f, \nu_\ell^\emptyset(t) \rangle \rightarrow \langle f, \bar{\nu}_\ell \rangle \quad (t \rightarrow \infty).$$

Because ℓ and f were arbitrary, this is exactly product weak convergence. $\square$

Lemma 3 (Order transfer from the prelimit). *Assume the FIFO+DHR hypotheses of Bramson–Lu–Prabhakar, and let H_0^1, H_0^2 be probability laws on $\mathcal{Q}$ with*

$$H_0^1 \preceq H_0^2$$

in the queue-state stochastic order induced by $0 \preceq (k, a)$ and $(k, a) \preceq (k', a') \iff k \leq k', a \leq a'$.

Suppose the hydrodynamic limit theorem applies to i.i.d. initial queue states with laws H_0^1 and H_0^2 , producing hydrodynamic solutions $\nu^1(\cdot)$ and $\nu^2(\cdot)$ with initial data $T(H_0^1)$ and $T(H_0^2)$. Then for every $\ell \geq 1$, every $t \geq 0$, and every bounded nondecreasing continuous $f : [0, L] \rightarrow \mathbb{R}$,

$$\langle f, \nu_\ell^1(t) \rangle \leq \langle f, \nu_\ell^2(t) \rangle. \quad (4)$$

Consequently, for every $x \in [0, L]$,

$$\nu_\ell^1(t)([x, \infty)) \leq \nu_\ell^2(t)([x, \infty)). \quad (5)$$

Proof. By Strassen's theorem, there exists a coupling $(Y_i^1(0), Y_i^2(0))_{i=1}^N$ of i.i.d. initial queue states with laws H_0^1, H_0^2 such that

$$Y_i^1(0) \preceq Y_i^2(0) \quad \text{a.s. for every } i.$$

Run the two N -server systems under Bramson–Lu–Prabhakar's standard coupling. Their Lemma 4.1 states that the queuewise preorder is preserved pathwise for all $t \geq 0$. Hypothesis audit: their lemma requires FIFO, decreasing hazard rate, and queuewise initial order in the (z, a) -coordinates; those are exactly the present assumptions. (web.stanford.edu)

Hence, for every bounded nondecreasing continuous f , the observable $\Phi_{\ell, f}$ is increasing on $\mathcal{Q}$, so

$$\mathbb{E}[\Phi_{\ell, f}(X_1^{N,1}(t))] \leq \mathbb{E}[\Phi_{\ell, f}(X_1^{N,2}(t))]. \quad (6)$$

As in the previous lemma,

$$\mathbb{E}[\Phi_{\ell, f}(X_1^{N,i}(t))] = \mathbb{E}\langle f, \nu_\ell^{(N),i}(t) \rangle.$$

Passing $N \rightarrow \infty$ by the hydrodynamic limit theorem gives (4).

To obtain (5), choose a standard sequence $f_m \in C_b([0, L])$ of nondecreasing continuous functions with $0 \leq f_m \uparrow \mathbf{1}_{[x, \infty)}$. Applying (4) to f_m and sending $m \rightarrow \infty$ yields (5) by monotone convergence. $\square$

I now prove the theorem.

Proof of Theorem. Step 1: identify the equilibrium environment with the physical invariant state.

In the Bramson–Lu–Prabhakar regimes, let E^∞ be the unique equilibrium environment. The associated $\bar{\nu} := T(E^\infty)$ is invariant for the mean-field/hydrodynamic evolution by definition of an equilibrium environment. Because the problem assumes uniqueness of the physical invariant state, it follows that

$$\bar{\nu} = \nu^*.$$

This is the target state below.

Step 2: prove empty-start convergence in the two regimes.

FIFO+DHR regime. Bramson–Lu–Prabhakar Theorem 2.2 states that, for FIFO SQ(d) with decreasing hazard rate and $\lambda < 1$, the finite- N equilibrium marginals converge to a unique equilibrium environment E^∞ . Their Section 5 further proves N -uniform relaxation from the empty state to equilibrium; the prose immediately preceding Proposition 5.2 makes explicit that, after some time q_1 independent of N , the tagged queue in the empty-start process and in the equilibrium-start process coincide throughout a whole interval $[q, q + T]$ outside an event of probability at most γ_1 . Hypothesis audit: their theorem and coupling proposition require fixed d , FIFO, DHR, and $\lambda < 1$; these are exactly the present assumptions. (web.stanford.edu)

Thus the previous lemma applies with $\bar{\nu} = \nu^*$, and yields

$$\nu^\emptyset(t) \rightarrow \nu^*. \quad (7)$$

Small-load regime. Bramson–Lu–Prabhakar Theorem 2.3 states that, for a generalized supermarket model with finite second moment θ and

$$\lambda \leq \frac{1}{2\sqrt{d(d \vee \theta)}},$$

the finite- N equilibrium marginals converge to a unique equilibrium environment E^∞ . Their Proposition 6.1 is the corresponding N -uniform empty-to-equilibrium relaxation statement. Hypothesis audit: SQ(d) is a generalized supermarket model in their sense, FIFO is a local service discipline, and the present moment/load assumptions match theirs exactly. (web.stanford.edu)

Therefore the empty-start convergence principle again gives

$$\nu^\emptyset(t) \rightarrow \nu^*. \quad (8)$$

This proves part (II).

Step 3: extend the FIFO+DHR result to the order interval $\delta_0 \preceq H_0 \preceq E^\infty$.

Assume now FIFO+DHR and let H_0 satisfy

$$\delta_0 \preceq H_0 \preceq E^\infty.$$

Let $\nu(\cdot)$ be the hydrodynamic solution with initial condition $T(H_0)$.

Apply the order-transfer lemma twice:

$$\delta_0 \preceq H_0 \preceq E^\infty \implies \nu_\ell^\emptyset(t)([x, \infty)) \leq \nu_\ell(t)([x, \infty)) \leq \nu_\ell^*([x, \infty)) \quad (9)$$

for every $\ell \geq 1$, every $x \in [0, L)$, and every $t \geq 0$.

By (7), $\nu_\ell^\emptyset(t) \Rightarrow \nu_\ell^*$ for each ℓ . Since the invariant-state paper shows that every physical invariant measure has the density form

$$\nu_\ell^*(dx) = r_\ell(x) \bar{G}(x) dx,$$

the measure ν_ℓ^* has no atoms. Hypothesis audit: the prompt already assumes precisely the invariant-state framework of Agarwal–Ramanan, and their Theorem 1 gives the density representation. (par.nsf.gov)

Fix ℓ and x . Because ν_ℓ^* is atomless,

$$\nu_\ell^\emptyset(t)([x, \infty)) \rightarrow \nu_\ell^*([x, \infty)).$$

Together with (9), the squeeze theorem gives

$$\nu_\ell(t)([x, \infty)) \rightarrow \nu_\ell^*([x, \infty)) \quad \text{for every } x \in [0, L). \quad (10)$$

For finite Borel measures on $[0, L)$, convergence of all tail functions to an atomless limit is equivalent to weak convergence. Therefore

$$\nu_\ell(t) \Rightarrow \nu_\ell^* \quad (t \rightarrow \infty)$$

for every $\ell \geq 1$. Equivalently,

$$\langle f, \nu_\ell(t) \rangle \rightarrow \langle f, \nu_\ell^* \rangle \quad \forall \ell \geq 1, \forall f \in C_b([0, L)).$$

This is precisely convergence in product weak topology, proving part (I). $\square$

Gap to the full statement

What remains open is the genuinely global theorem suggested by the prompt:

for *every* physically relevant initial condition $\nu(0)$ in a large finite-mean class, and for *general* service distributions covered by the hydrodynamic equations, prove $\nu(t) \rightarrow \nu^*$.

My proof does *not* cover:

- arbitrary initial conditions in the general well-posedness class;
- arbitrary initial conditions even in the FIFO+DHR regime, unless they lie in the stochastic order interval $[\delta_0, E^\infty]$;
- nonempty initial conditions in the small-load regime without an additional comparison principle.

Moreover, as the 2025 invariant-state paper explicitly says, proving convergence of hydrodynamic solutions to the invariant state for a large class of initial conditions was left for future work. (par.nsf.gov)

Self red-team

- **Counterexample attempt to the strongest reading.** The strongest reading “for all $\nu(0) \in \mathcal{S}$ ” is false. Indeed the nonphysical state $\nu_\ell(dx) = \bar{G}(x)dx$ for every ℓ is invariant, so a trajectory started there never approaches the physical invariant state ν^* . This is why I restricted to a physical subclass. (par.nsf.gov)
- **Referee objection 1: does E^∞ really correspond to ν^* ?** I addressed this by defining $\bar{\nu} := T(E^\infty)$, noting that E^∞ is an equilibrium environment (hence induces an invariant mean-field law), and then using the assumed uniqueness of the physical invariant state to identify $\bar{\nu} = \nu^*$.
- **Referee objection 2: Proposition 5.2 / 6.1 is OCR-garbled.** True: the displayed inequalities in the OCR output lose a complement. I therefore audited the surrounding prose, which states the intended meaning explicitly: after large time q_1 independent of N , the tagged queues of the empty-start and equilibrium-start systems agree throughout $[q, q+T]$ off a negligible event. That stronger prose statement is exactly what I use. (web.stanford.edu)
- **Referee objection 3: do tail inequalities imply weak convergence?** Yes, because the limit ν_ℓ^* is atomless in the age coordinate. For finite measures on $[0, L)$, convergence of $\mu_t([x, \infty))$ at every x to an atomless limit measure is equivalent to weak convergence.
- **Boundary checks.**
 - At $\nu(0) = \nu^*$, the theorem is consistent: the solution is constant.
 - At $\nu(0) = 0$, part (I) reduces to part (II) when both regimes hold.
 - The theorem does not claim anything at $\lambda = 1$, where both the prompt and the cited queueing results change character.

Ideas tried

1. **Direct Lyapunov on the hydrodynamic equations.** I tried to build a workload-type Lyapunov functional from the balance identities

$$\langle 1, \nu_\ell(t) \rangle = \langle 1, \nu_\ell(0) \rangle - D_\ell(t) + D_{\ell+1}(t) + \int_0^t \langle 1, \eta_\ell(s) \rangle ds,$$

and from telescoping sums over ℓ . This yields useful identities, but I could not extract a coercive monotone functional for the full nonlinear measure-valued system.

2. **ω -limit set plus uniqueness of the invariant state.** I explored proving precompactness of the orbit and then identifying every ω -limit point as invariant. The obstruction is that uniqueness of the invariant state alone does not rule out more complicated invariant sets (e.g. periodic orbits), and I did not obtain a separate Lyapunov functional that collapses the whole ω -limit set to a singleton.
3. **Monotone semiflow / Hirsch theory.** For exponential or phase-type reductions, one can hope for a cooperative ODE and a monotone dynamical-systems theorem. But for the present general-service measure-valued equations, I could not verify the strong-order or compactness hypotheses needed for a clean application. The only robust monotonicity I found on the web is Bramson–Lu–Prabhakar’s FIFO+DHR queue-state monotonicity, and I used exactly that in the successful proof.
4. **Successful approach: pull stability from the prelimit.** The decisive step was to combine:
 - (a) N -uniform relaxation of the finite- N tagged queue from empty to equilibrium,
 - (b) convergence of finite- N equilibrium marginals to a unique equilibrium environment,
 - (c) the hydrodynamic limit.

This yields empty-start convergence immediately; in the DHR regime, prelimit monotonicity then upgrades it to a whole stochastic order interval of initial data.

Invoked results

1. **Hydrodynamic limit for SQ(d) (T00L).** *Statement used.* Under the standing regularity assumptions on G and for exchangeable initial conditions whose scaled measure-valued state converges to $\nu(0) \in \mathcal{S}$, the scaled state process $\nu^{(N)}(\cdot)$ converges on every finite time interval to the unique solution $\nu(\cdot)$ of the hydrodynamic equations with initial condition $\nu(0)$. (arxiv.org)
Hypothesis audit. The prompt assumes exactly the well-posedness regime in which the hydrodynamic equations have unique solutions. My initial data are i.i.d. across queues (hence exchangeable) and have deterministic law-of-large-numbers limits $T(H_0)$, so the theorem applies.
2. **Bramson–Lu–Prabhakar Theorem 2.2 and Proposition 5.2 (T00L).** *Statement used.* For FIFO SQ(d) with decreasing hazard rate and $\lambda < 1$:
 - the finite- N equilibrium marginals converge to a unique equilibrium environment E^∞ ;

- the tagged queue in the empty-start process relaxes to the tagged queue in the equilibrium-start process uniformly in N , in the sense explained in the prose around Proposition 5.2. (web.stanford.edu)

Hypothesis audit. In part (I), the model is exactly FIFO SQ(d), d is fixed, $\lambda \in (0, 1)$, and h is decreasing. Thus the result applies verbatim.

3. **Bramson–Lu–Prabhakar Theorem 2.3 and Proposition 6.1 (T00L).** *Statement used.* For a generalized supermarket model with fixed d , finite service-time second moment θ , and

$$\lambda \leq \frac{1}{2\sqrt{d(d \vee \theta)}},$$

the finite- N equilibrium marginals converge to a unique equilibrium environment E^∞ , and the empty-start tagged queue relaxes to equilibrium uniformly in N . (web.stanford.edu)

Hypothesis audit. SQ(d) is a generalized supermarket model in their sense; FIFO is local; I explicitly assume the same moment and load bound. Hence the theorem applies.

4. **Bramson–Lu–Prabhakar Lemma 4.1 (T00L).** *Statement used.* Under the standard coupling for FIFO SQ(d) with decreasing hazard rate, the queuewise preorder

$$(z, a) \preceq (z', a') \iff z \leq z', a \leq a'$$

is preserved pathwise for all times. (web.stanford.edu)

Hypothesis audit. In part (I), I work exactly in the FIFO+DHR regime, and I choose the initial queuewise coupling by Strassen so the initial preorder holds almost surely. Therefore the lemma applies.

5. **Agarwal–Ramanan Theorem 1 / density form of invariant states (T00L).** *Statement used.* A physical invariant state has the density representation

$$\nu_\ell^*(dx) = r_\ell(x)\bar{G}(x) dx,$$

so each ν_ℓ^* is atomless in age. (par.nsf.gov)

Hypothesis audit. The prompt assumes the regularity conditions used in that paper and the existence/uniqueness of the physical invariant state. I use only the absolute continuity conclusion.

Self-confidence

3 — The partial theorem itself is tightly argued, but it relies on several external queueing-limit theorems and on one OCR-corrected reading of a proposition statement, so a hidden mismatch in those imported hypotheses remains possible.